

E-Commerce Sales Performance Analysis

Business Insights Report

Dataset: Sample Superstore (2014–2017)
Orders Analyzed: 5,009
Tools Used: SQL (SQLite), Python (Pandas, Matplotlib), Google Looker Studio

Executive Summary

Headline Metrics

Metric	Value
Total Revenue	\$2,297,200.86
Total Profit	\$286,397.02
Profit Margin	12.47%
Total Orders	5,009

The business generated approximately **\$2.3 million in revenue** between 2014 and 2017 while maintaining a **12.47% profit margin**. Compared with a target margin of 20%, the company is operating roughly **7.5 percentage points below its potential profitability**, representing approximately **\$172,000 in unrealized profit** over the period analyzed.

Revenue increased steadily year over year, with a recurring surge during the fourth quarter driven by seasonal demand. Monthly sales peaked at approximately **\$120,000 in November 2017**, indicating strong customer demand. The primary challenge is therefore not revenue generation, but profit retention.

Revenue Trend Analysis

Revenue growth remained consistently positive throughout the four-year period. The seasonal pattern is highly predictable, with sales accelerating between October and December every year.

This recurring trend provides a clear operational insight: inventory planning, staffing levels, and logistics capacity should be scaled proactively before every Q4 peak period.

Despite growing revenue, profit margins remained relatively flat. Revenue growth without margin improvement suggests a structural profitability issue, most likely linked to pricing and discounting practices rather than customer demand.

Detailed Findings

Finding 1: Category Profitability

Category	Revenue	Profit	Margin
Technology	\$330,007	\$44,516	13.49%
Office Supplies	\$719,047	\$122,491	17.04%
Furniture	\$742,000	\$60,299	8.12%

Office Supplies generated the highest profit margin at **17.04%**, supported by highly profitable sub-categories such as Paper, Labels, and Art.

Furniture generated the highest revenue but produced only an **8.12% margin**, less than half that of Office Supplies. While the category contributes significantly to sales volume, it underperforms in profitability and acts as the largest drag on overall company margins.

Technology remains a strong performer, balancing healthy revenue generation with consistent profitability.

Key Insight

The company's profitability challenge is concentrated within the Furniture category rather than spread evenly across the business.

Finding 2: Loss-Making Sub-Categories

Sub-Category	Revenue	Profit	Margin
Tables	\$206,965	-\$17,725	-8.56%
Bookcases	\$114,880	-\$3,473	-3.02%
Supplies	\$46,674	-\$1,189	-2.55%
Fasteners	\$3,024	-\$950	-31.40%

Tables is the most significant loss-making sub-category in the portfolio. Despite generating over **\$206,000 in revenue**, it produced a loss of **\$17,725**, indicating that high sales volume alone does not guarantee profitability.

The issue is particularly concerning because Tables represent a major share of category sales. This is not an isolated product problem but a high-volume profitability problem.

In contrast, several sub-categories demonstrate exceptionally strong margins:

Sub-Category	Revenue	Profit	Margin
Paper	\$78,479	\$34,054	43.39%
Labels	\$12,486	\$5,546	44.42%
Art	\$16,476	\$6,964	42.27%

The same business generating margins above 40% on Paper and Labels is simultaneously losing money on Tables, suggesting that pricing and discounting policies—not market demand—are driving the performance gap.

Key Insight

Losses are concentrated in a small number of heavily discounted, high-volume Furniture products.

Finding 3: Geographic Performance

Profitability is concentrated in a handful of high-performing states.

Top-performing states include:

- California (~\$75K profit)

- New York (~\$73K profit)
- Washington
- Michigan
- Virginia
- Indiana
- Georgia
- Kentucky
- Minnesota

The geographic profit map reveals that while most markets remain profitable, several cities generate recurring losses.

One notable exception is Texas, which generated only **\$20.22 profit** despite being one of the largest retail markets in the United States. This effectively represents break-even performance and indicates potential pricing, discounting, or operational inefficiencies within the state.

Key Insight

Texas represents a significant opportunity for margin improvement and warrants further investigation.

Finding 4: Customer Segmentation (RFM Analysis)

The RFM model segmented 793 customers into six behavioral groups.

Customer Distribution

Segment	Customers	Share
Potential Loyalists	210	26.5%
Loyal Customers	165	20.8%
At Risk	158	19.9%
Lost	113	14.2%
New Customers	74	9.3%
Champions	73	9.2%

Revenue by Segment

Segment	Customers	Avg Days Since Purchase	Avg Orders	Revenue
Loyal Customers	165	39.5	8.01	\$610,591
At Risk	158	180.67	8.08	\$600,839
Potential Loyalists	210	141.55	4.55	\$542,429
Champions	73	17.23	9.74	\$291,241
New Customers	74	16.11	4.55	\$141,781
Lost	113	442.20	3.61	\$110,320

The most valuable insight is the similarity between the **At Risk** and **Loyal Customer** segments.

At Risk customers have generated nearly identical revenue and order frequency compared with Loyal Customers. The primary difference is recency of purchase.

These customers are not low-value buyers; they are previously engaged customers who have become inactive.

Key Insight

The At Risk segment represents the largest recoverable revenue opportunity in the customer base.

Recommendations

Recommendation 1: Implement a 20% Discount Cap on Furniture Products

Evidence

Tables (-8.56% margin) and Bookcases (-3.02% margin) are among the most heavily discounted products and consistently generate losses.

Action

- Implement a maximum discount threshold of 20% for Furniture products.

- Require management approval for any exception.
- Align sales incentives with profitability rather than revenue volume.

Expected Impact

Recovering losses from Tables, Bookcases, and Supplies could generate approximately **\$22,000–25,000 in additional profit** based on historical performance.

Recommendation 2: Launch a Win-Back Campaign for At Risk Customers

Evidence

The At Risk segment generated over **\$600,000 in revenue** and averages more than eight orders per customer, matching the behavior of Loyal Customers.

Action

- Export the At Risk segment from the RFM dataset.
- Send personalized email campaigns referencing previous purchases.
- Offer targeted loyalty incentives such as a 10% discount on preferred product categories.

Expected Impact

Assuming a conservative 25% reactivation rate, approximately 40 customers could be recovered, generating an estimated **\$150,000–180,000 in recovered revenue**.

Recommendation 3: Shift Marketing Spend Toward High-Margin Products

Evidence

Paper, Labels, Art, and Copiers consistently generate strong margins ranging from 25% to 44%.

Action

- Reallocate 20% of Furniture promotional spending toward high-margin Office Supplies and Technology products.
- Build bundled offers for corporate customers.
- Reduce promotional activity on Tables and Bookcases.

Expected Impact

A 3–5 percentage point improvement in promotional mix margin could generate approximately **\$35,000–55,000 in additional profit** at current revenue levels.

Strategic Action Summary

Recommendation	Priority	Timeline	Estimated Impact
Cap Furniture Discounts at 20%	High	Immediate	+\$22K–25K Profit
Win-Back Campaign for At Risk Customers	High	2–4 Weeks	\$150K–180K Revenue Recovery
Promote High-Margin Products	Medium	1 Quarter	+\$35K–55K Profit

Conclusion

The analysis shows that the business does not suffer from a revenue-generation problem. Demand is strong, revenue is growing, and customer engagement remains healthy.

The primary challenge is profitability.

A small number of heavily discounted Furniture products are eroding margins, while a large pool of high-value customers remains underutilized. By tightening discount controls, reactivating At Risk customers, and prioritizing high-margin products in marketing campaigns, the business could recover or protect an estimated **\$200,000–260,000 in value over the next 12 months**.

These opportunities are supported directly by observed sales, profit, customer, and geographic performance data, making them actionable, measurable, and commercially realistic.